CLOUD-INTEGRATED IOT AND ML-BASED WASTE MANAGEMENT SYSTEM FOR RAILWAY STATIONS

25-26J-191

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: 24/8/25 Date:

Abstract

Efficient waste management in high-traffic public spaces continues to be a significant challenge, particularly in railway stations where trash buildup is highly unpredictable and variable depending on commuter traffic. Conventional approaches continue to rely on manual labor and set collection schedules that more often than not result in overflowing bins, bad smells, and overall inefficiency. All these disadvantages point towards the need for intelligent automated systems capable of dynamically reacting to real-life scenarios with minimum human effort. Responding to this need, this study puts forward a cloud-supported autonomous waste collection system capable of detecting, collecting, transporting, and unloading the trash in railway stations. By integrating robotics, IoT-enabled sensing, and intelligent control systems, the system is designed to provide round-the-clock cleanliness, minimize operational downtime, and enhance the overall commuter experience. At its core is the robotic intelligent navigation and unloading functionality. The robot navigates autonomously in complex station platform designs, avoiding static and dynamic obstacles. Height-compensated ultrasonic sensors provide multilevel obstacle detection, and edge detection provides accidental drop protection at platform edges. In addition, the system incorporates blueprint-based spatial awareness, which enables the robot to sense predefined safe zones, static obstacles and platform edges. This is particularly useful in high crowd density levels, as the robot can navigate to the nearest safe zone and power down temporarily until crowd congestion decreases. An internal bin monitoring system initiates unloading only when the waste level exceeds a set limit, and GPS-based location awareness guides the robot to designated disposal sites to unload safely. Integrating obstacle detection, blueprint-based safety planning, bin-level monitoring, autonomous unloading control, and cloud-enabled data integration, this solution provides a very flexible and railway-specific solution to automated waste management with enhanced safety, reliability, and efficiency.

Keywords: Autonomous navigation, Blueprint-based safety planning, Autonomous unloading control, Obstacle avoidance, Bin-level monitoring

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
IoT	Internet of Things
AI	Artificial Intelligence
SLAM	Simultaneous Localization and Mapping
IR Sensor	Infrared Sensor
GPS	Global Positioning System
IMU	Inertial Measurement Unit
QA	Quality Assurance
IDE	Integrated Development Environment
MS Planner	Microsoft Planner

1. INTRODUCTION

1.1 Background & Literature Review

Efficient waste management in railway stations is a matter of utmost concern over the past few years with growing dependence on public transport systems and the constant flow of passenger flow in both urban and suburban areas. The random disposal tendency of passengers has the effect of scattering litter across platforms, tracks, and waiting rooms, creating hygiene problems, safety risks, and increasing burdens on the cleaning staff. Conventional waste disposal mechanisms rely on fixed bins and manual cleaning operations, which are inherently retrospective in nature and have a tendency to overlook waste accumulation in real time. Overflowing bins, litter left uncollected for extended durations, and delays in cleaning not only decrease passenger satisfaction but also impact operating efficiency. Such limitations have fostered an increased demand for autonomous robotic systems [1], powered by IoT and AI [2], that can sense, collect, and remove the waste adaptively in these dynamic environments.

Mobile waste-collecting robot navigation is a variety of technical challenges for railway stations. They are high-pedestrian-density semi-structured zones with temporary obstacles such as baggage, and presence of high-level platforms that are significant safety hazards. There have been traditional approaches to crossing mobile robots that depend primarily on mapping-based methods such as Simultaneous Localization and Mapping (SLAM) [3] together with global path-planning methods such as Dijkstra or A* [4]. While such strategies are sufficient for controlled or constant environments, they do not apply as well to busy stations where the environment is in flux due to passenger traffic and mobile obstacles. Such a constraint has driven scholars to explore sensor-based, reactive navigation strategies where ultrasonic and proximity sensors allow robots to detect obstacles in real time and adjust their motion accordingly [5]. Such reactive navigation avoids computation overhead of SLAM-based methods with robots remaining agile and adaptable in open environments.

In autonomous navigation research, the use of ultrasonic sensors has been particularly emphasized due to their cost-effectiveness, resilience to dynamically varying lights, and effective short-range sensing. Experiments demonstrate that the use of ultrasonic sensors at various heights enhances environmental perception [6] in that robots can differentiate between litter at floor level and large sized objects such as benches or human legs [7]. This feature is especially relevant in railway stations, where litter is scattered and individuals occupy the same area. Cliff detection systems also play a critical role in safety guarantee. Higher platforms in train stations require robots to sense drop-offs and not fall, which would damage trains or disturb train movement. Research on cleaning and service robots indicates that downward-looking IR and proximity sensors [8], [9] function best to sense unexpected elevation changes and provide cheap but effective protection against falling. These comments imply that the navigation system of a railway station must possess fall prevention as well as obstacle avoidance functions to ensure a secure and reliable operation.

Besides navigation, effective waste management robots must also identify dumping sites automatically and off-load waste collected. The majority of current research on indoor service robots inevitably relies on reference markers, QR codes, or Wi-Fi triangulation to find unique dumping spots [10]. Nevertheless, train stations usually span large, partly-covered, or open spaces, whereby such methods prove to be infeasible. Global Positioning System (GPS) technology augmented by sensor fusion with odometry and inertial measurement units (IMUs) provides a more viable method in such cases. Conventional GPS loses accuracy in partially screened environments, but new advancements in RTK-GPS and PPP-RTK [11], [12] achieved sub-meter precision in robotic applications. Experiments confirm that GPS aided with ultrasonic alignment and IMUs allows robots to precisely deliver recyclables to fixed disposal sites even in challenging conditions [13]. This means unloading strategies with locational capability can ensure robust performance in outdoor and semi-indoor conditions of railway stations.

Level monitoring of bins is another critical dimension of managing unloading. Athe robot must not unload waste unnecessarily, wasting energy and time, or keep unnecessary waiting until an overflow state. Ultrasonic sensor-based internal bin level monitoring has effectively been employed in IoT-capable waste management systems to determine fill levels [14] and only initiate forced collection or disposal when required. Through the integration of such sensors in the onboard bin of the robot, waste collection is optimized and responsive. Research by Srivastava et al. [8] indicates the potential use of IoT-based bin monitoring towards enhancing collection route optimization and preventing overflows. Through integration into cloud systems, this monitoring allows operators to see real-time bin conditions and receive alerts for bins requiring emptying [15], [16]. Utilizing such approaches within mobile robots closes the decision loop whereby bin levels will indeed trigger unloading behavior, improving real-world deployment operational efficiency.

Despite progress being made on autonomous navigation, bin monitoring, obstacle detection, and unloading control, these are primarily explored in isolation rather than as an integrated system. Autonomous waste robots such as AutoClean [11] demonstrate the potential of mobile waste robots but are focused on generic environments such as city roads or campuses rather than the particular requirements of railway stations. Similarly, smart bins based on IoT have advanced in monitoring and alerting features [14], but they are stationary in nature and fail to solve the problem of littered waste in densely populated areas. Even shopping mall and airport service robots [6], [10] reference autonomy potential without resolving raised platform and safety issues unique to rail facility transportation. Current systems usually use pre-scheduled timetables or human intervention to unload, in contrast to condition-based decisions made from real-time monitoring of bins.

This literature review refers to a significant area of study: though each of the stand-alone technologies for navigation, obstacle avoidance, falling object detection, bin monitoring, and unloading has been implemented, not much has attempted to place them in a system-wide mode specific to railway stations. The challenge is to embed reactive sensor-based navigation with safety-critical functions such as platform-edge detection without compromising on smart unloading decisions enabled by real-time bin monitoring and reliable disposal station localization. This gap is not only an academic breakthrough in autonomous robots but also a practical leap towards waste management sustainability in public transport systems. By bringing these technologies together into a single platform, waste-collecting robots are able to operate in railway stations' extreme and volatile environments with minimal dependence on humans, thus enhancing cleanliness and passenger experience.

1.2 Research Gap

Automated waste collection by mobile robots has gained growing interest in the last years, and experimental systems have demonstrated autonomous collection and navigation in controlled or semi-structured environments [1], [5]. The majority of existing approaches, however, rely on predefined routes [4], static maps, or SLAM-based approaches [3], which are effective in structured indoor facilities but not adaptable to semi-structured and dynamic environments such as railway stations. These environments present unique challenges that consist of dynamic pedestrian density, non-homogeneous waste distribution, and the presence of high platforms with high robot fall hazards. Although research in navigation has advanced, current work has not adequately addressed these safety-critical conditions, where preventing platform drop-offs [9] is equally as important as obstacle avoidance [10].

A second important gap is the integration of unloading mechanisms with navigation control. Although garbage-collecting robots exist, their unloading tasks are usually semi-automatic or human-assisted, sacrificing total system autonomy [11]. Docking solutions through reference markers or QR codes have been proposed [10] but require physical environment changes, which are infeasible in high-traffic public transportation zones. GPS localization was successfully implemented in outdoor service robots but remains less investigated in railway stations, particularly in semi-covered platforms where the signals are weakened. Recent studies demonstrate that GPS fusion with inertial sensors, odometry, and short-range ultrasonic alignment can provide more robust disposal station detection [12], [13]. The hybrid approach is still to be sufficiently implemented on mobile waste-collecting robots in railway stations.

The third and equally important gap is the lack of integration between bin-level monitoring and unloading control. While stationary smart bins equipped with ultrasonic sensors and IoT connectivity [2] are widely studied for fill-level monitoring and cloud-based notifications [8], [16], few works extend this concept to mobile robots. As a result, most robots unload waste either prematurely, with wastage of resources, or too late, with a danger of overfilling. Lacking real-time integration of internal bin status into navigation and unloading decision-making, current systems lack the intelligence to facilitate efficient operations. In addition, most current research addresses navigation [6], unloading, and bin monitoring [15] separately, rather than integrating them in a common framework that can facilitate real-time decision-making in high-traffic, safety-critical railway networks.

To demonstrate these limitations, *Table 1.1* presents a systematic comparison of existing systems with the requirements of railway stations: (i) safe obstacle and drop-off prevention, (ii) intelligent unloading without environment adaptation, and (iii) fill-level–triggered unloading with supervisory monitoring integration. The comparison shows that existing solutions still remain fragmented, addressing elements of the problem but not an integrated, autonomous system expressly conceived for railway station environments. These gaps strongly inspire the quest to develop intelligent navigation and unloading control in the context of mobile waste-collecting robots.

Table 1.1 Comparison of previous researches

Aspect	Existing Systems	Research Gaps Identified
Navigation & Obstacle Avoidance	SLAM-based mapping, predefined routes, ultrasonic/IR sensors for object detection and collision avoidance.	Effective in structured indoor settings but inadequate for semi-structured, dynamic environments like railway stations; insufficient attention to drop-off (platform edge) prevention
Platform Safety (Drop-off Detection)	IR and ToF sensing for cliff detection in domestic/service robots.	Not adapted for railway platforms with higher safety risks and crowded conditions
Unloading Mechanisms	Semi-automatic unloading or human-triggered disposal; docking via reference markers/QR codes.	Requires environmental modifications (markers/codes), reducing practicality in busy transport hubs; limited autonomy

Localization for Disposal	GPS-based positioning used in outdoor robots; recent improvements with sensor fusion (GPS + IMU + odometry)	GPS degrades in semi-covered railway platforms; hybrid solutions underexplored in waste-collection robots
Bin-Level Monitoring	IoT-enabled static bins with ultrasonic fill sensors and cloud alerts	Mostly limited to stationary bins; rarely integrated with mobile robots to guide autonomous unloading decisions
Integrated Framework (Navigation + Unloading + Monitoring)	Individual studies on navigation, unloading, or monitoring separately	Very limited research combining all three into a unified, adaptive solution for railway stations

1.3 Research Problem

How to architect autonomous mobile robots that can strongly navigate semi-structured, safety-critical environments such as railway stations, steer clear of collisions, guard against falls from platforms, and locate disposal points exactly with GPS-based location awareness, in order to support completely automatic waste unloading?

Railway stations present this challenge because they are extremely dynamic environments with fluctuating pedestrian density, unexpected obstacles, and elevated platforms, which provide unique safety hazards. Existing navigation technologies in robots are primarily calibrated for indoor controlled environments or campus-style outdoor environments, where mapping, localization, and obstacle avoidance can be solved using high-level algorithms such as SLAM or path planning. However, in collecting robots, where cheap, sensor-based solutions must execute in real time, robust navigation independent of computationally heavy frameworks remains a root issue.

Another side of the issue is ensuring correct unloading control after the internal bin of the robot has been filled. Although GPS and sensor fusion have been employed to improve robot localization, challenges continue in the semi-covered environment like railway stations where the signal quality degrades. Also, the discovery and driving to a definitive disposal location, alignment correctly, and control of an unloading system autonomously is an area under insufficient research.

Thus, the problem is not just about navigation in general robotics but about designing a cost-effective, reliable, and safe navigation-plus-unloading framework for waste-collecting robots in railway stations. The solution to the problem consists of bridging gaps between simple sensor-based avoidance, GPS-directed localization, and intelligent unloading decision-making in unstructured public environments.

2. OBJECTIVE

2.1 Main Objectives

The main objective is to develop a cloud-integrated, IoT- and machine-learning-based autonomous waste management system that can detect, collect, classify, and intelligently unload garbage in railway stations, while ensuring real-time monitoring of disposal bins and timely notifications for efficient waste handling.

2.2 Specific Objectives

- Autonomous Garbage Detection and Collection
    - To design and implement a vision-based detection mechanism supported by sensors and a robotic arm for identifying and collecting garbage items in real time.
- Intelligent Navigation and Unloading Control
    - To enable the mobile robot to autonomously navigate railway station environments with obstacle and fall prevention, continuously monitor internal bin fill levels, and perform automated unloading at predefined disposal areas.
- Real-Time Crowd Detection and Waste Classification
    - To develop and train custom machine learning models for classifying collected waste into four categories as Plastic, Paper, Glass, and Polythene at the disposal station, and detecting crowd density through head-count analysis using cameras to ensure safe navigation and operation.
- Automated Waste Sorting and Disposal Bin Monitoring
    - To integrate a robotic arm for placing classified waste into static bins and implement IoT-based monitoring of disposal bin fill levels, with cloud-based visualization and real-time notifications for cleaning staff.

3. METHODOLOGY

3.1 System Architecture

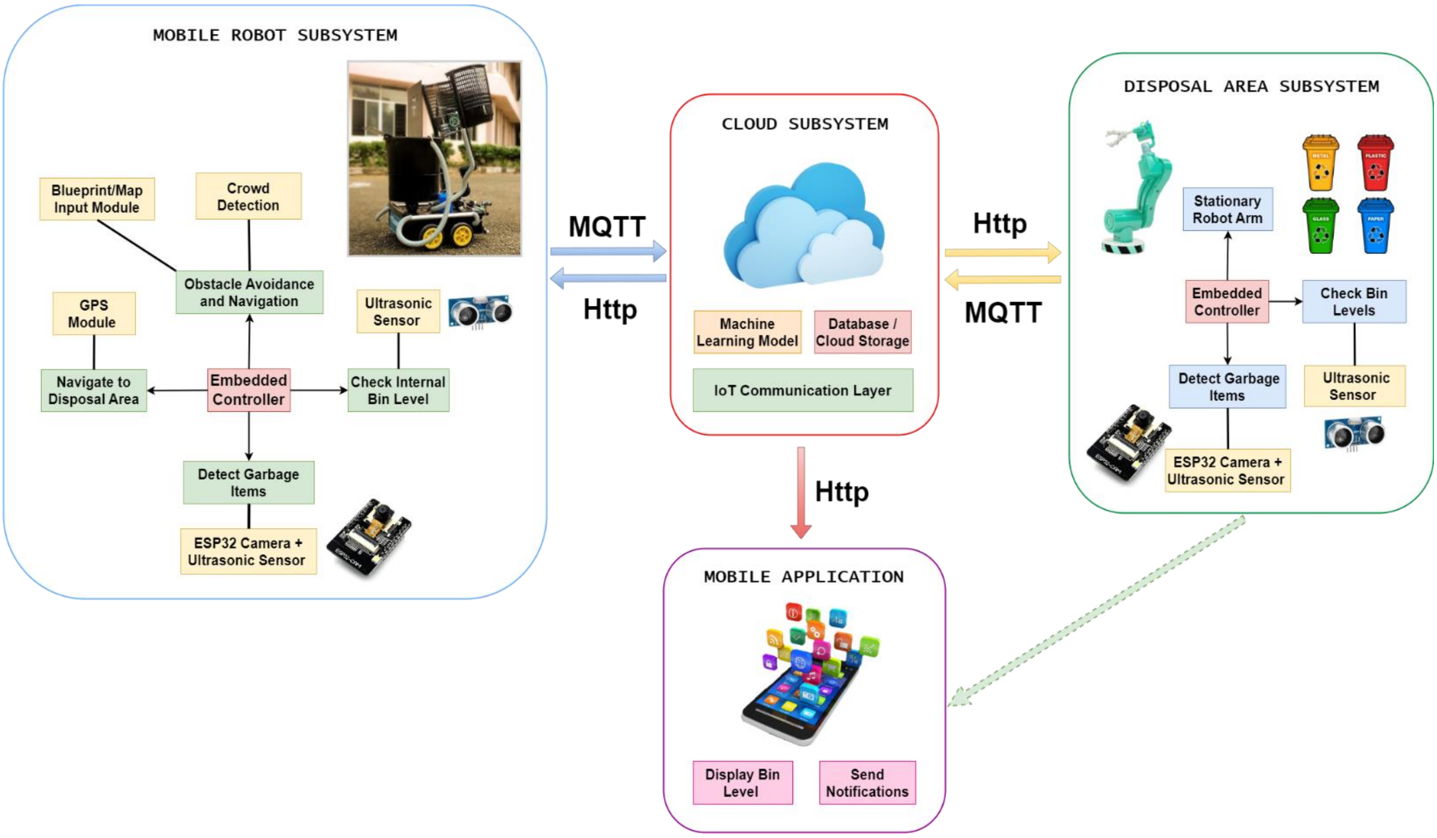

Figure 3.1: System architectural diagram for the overall system

Figure 3.1 presents the overall system architecture, illustrating how the four subsystems mobile robot, cloud, disposal area, and mobile application are interconnected to achieve seamless autonomous waste detection, collection, classification, and disposal in railway stations.

The system has been conceptualized as an integrated waste management system which utilizes robotics, IoT, and machine learning to make the railway stations cleaner. It has been segmented into four standalone subsystems that work in flawless synergy to create a fully autonomous, efficient, and reliable system for waste collection, segregation, and disposal. Each of the subsystems offers an individual set of capabilities, but how they interoperate is what results in the achievement of the overall research objectives.

- **Mobile Robot Subsystem**: This unit offers the essence of the solution by serving as the active unit for performing the process of waste collection. It has an onboard vision system supplemented by sensors that help identify garbage objects in real-time. Upon detection of an object, the robotic arm switches on automatically to pick up the trash and deposit it into the robot's internal storage compartment. The navigation module allows the robot to move freely through station environments without facing hazards and obstacles. Safety is ensured with the use of sensors placed at strategic positions to not only navigate objects around in its path, but also to avoid falling off elevated railway platforms. In addition, the subsystem integrates blueprint-based spatial awareness to pre-identify static obstacles, safe zones, and platform edges, enhancing navigation safety. The robot is also aided by a crowd detection mechanism via cameras and a learned model to stop navigating and move towards the closest predetermined safe spot upon detection of intense crowding, resuming operation once crowds have abated. The subsystem further possesses a vigilance system that checks the fill level of its internal tank constantly to offload only as needed. After filling, the robot moves automatically to the disposal point with the help of location awareness and GPS.
- **Cloud Subsystem**: The cloud subsystem provides central intelligence and connectivity. It is the site of the machine learning algorithm that translates picture data to waste classification to deliver accuracy in the discrimination between types such as plastic, glass, paper, and polythene. Apart from classification, the cloud gives the communications infrastructure, connecting the mobile robot, disposal site, and mobile application into a unified architecture. It also addresses data storage, with the ability to conduct long-term trend analysis of waste and scalability to scale the solution to numerous railway stations. Additionally, the cloud hosts the crowd detection model for head-count analysis, transmitting results to the robot in real time to guide safe navigation and trigger safe-zone protocols when necessary.

- **Disposal Area Subsystem**: In the disposal area subsystem, waste handling changes from collection to segregation. Once the mobile robot off-loads the internal bin onto the platform, it is passed over by a fixed robotic arm with inbuilt controllers. This arm, which is guided by machine learning classification output, sorts the collected items into dedicated bins. The disposal bins all feature ultrasonic sensors that constantly track and report fill levels. By embedding intelligence in this component of the system, the disposal is systematic and sustainable, reducing human interference and mistake in segregation of waste.
- **Mobile Application**: The mobile interface is the human-machine interface of the system. The disposal systems and the robot run independently, but the mobile app provides visibility and accountability. It displays actual fill levels for disposal bins, giving cleaning staff complete visibility of the system's status. Automatic notifications are issued when a bin is close to full capacity, prompting timely action. This ability closes the loop of autonomous and human monitoring, always remaining effective in protecting railway station sanitation.

Through the integration of the foregoing subsystems, the solution responds to several concerns simultaneously: autonomous garbage detection and pickup, free passage in crowded public areas, smart separation and sorting, and effective human-machine interaction. The combination of IoT, cloud-based intelligence, and robotics enables the railway stations to dynamically maintain cleanliness levels based on actual conditions rather than predetermined schedules. Coupling autonomy with intelligent notifications and real-time monitoring, the system offers a revolutionary approach to modern waste management of high-density public spaces.

3.2 Process Workflow

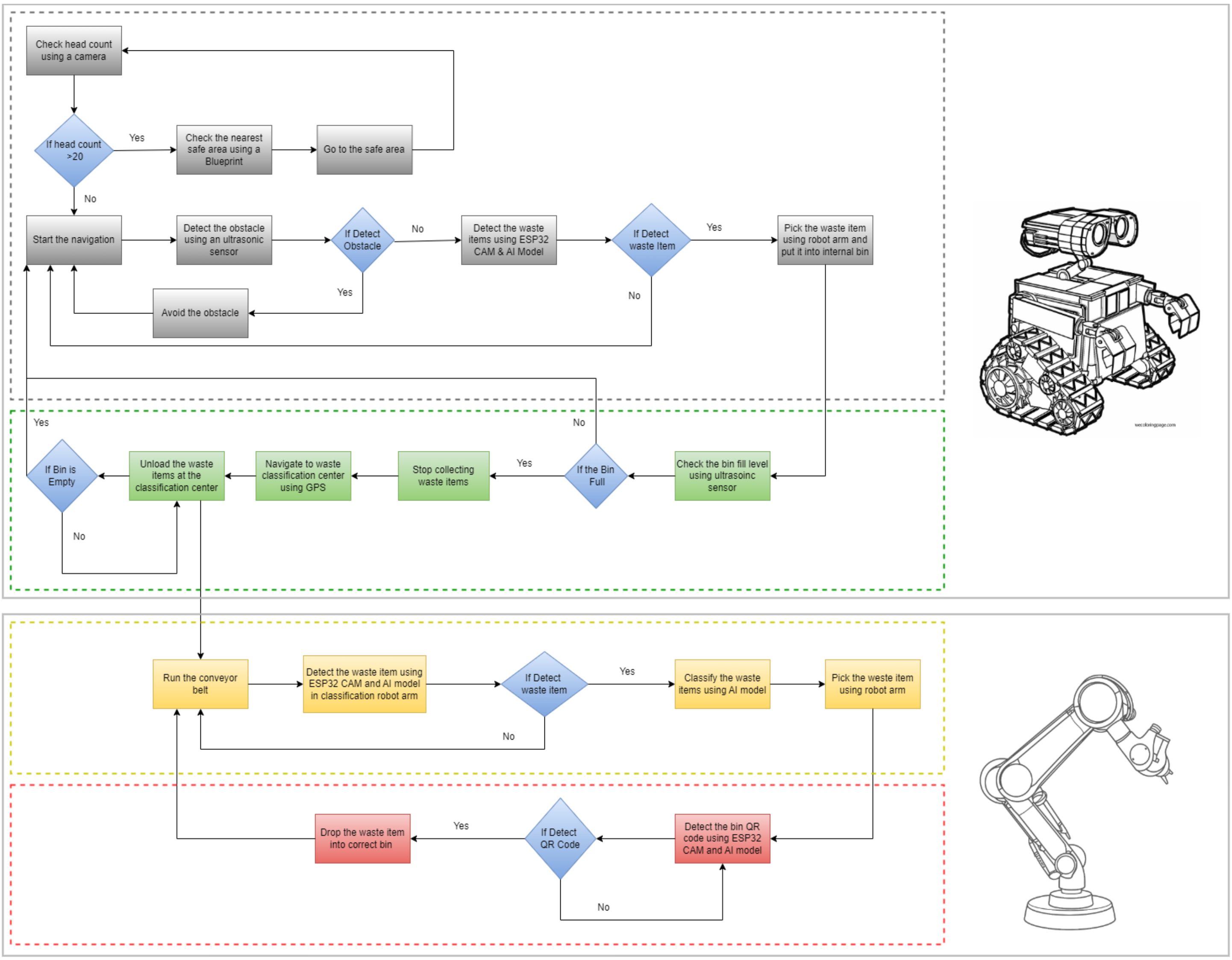

Figure 3.2: Flowchart for the overall system

The overall workflow of the proposed system has been designed to ensure that waste is identified, collected, transported, and classified autonomously and effectively with real-time monitoring and alert. The system begins with the autonomous robot actively navigating around the area to sense waste objects. According to onboard sensors and an in-built vision model, the robot separates potential garbage objects from other irrelevant environmental objects. After identifying a valid garbage object, the robotic arm is activated to grasp the object and deposit it into the robot's internal bin.

As the robot continues operation, the internal bin level is continuously monitored by ultrasonic sensing. When the bin is 80% (or a predetermined percentage) full, the system proceeds to the unloading phase. In this stage, the robot switches its task from active collection to navigation. The robot navigates to the disposal point via a combination of location awareness and obstacle avoidance so that the robot can safely operate within congested or semi-structured railway station environments. To enhance navigation, the robot also uses blueprint-based spatial awareness to pre-identify static obstacles, platform edges, and predefined safe zones. Furthermore, a crowd detection mechanism using camera input and a trained model monitors real-time head counts. If a crowded situation is detected during operation, the robot halts its collection task and moves to the nearest predefined safe zone, resuming only when congestion levels decrease.

Once the robot reaches the disposal area, it performs the unloading process, emptying collected waste onto a dumping stage. The disposal system is also triggered where the classification model deposits every item of waste into four bins named Plastic, Paper, Glass, and Metal. Each product is released by the fixed robotic arm into its designated dump bin, which has been presetup and placed in prearranged locations for standardization.

Simultaneously, all the garbage bins are equipped with ultrasonic sensors to constantly monitor their levels. These are sent via the IoT layer to a cloud database. A mobile application makes this visual, providing cleaning staff with real-time feedback and notifications when a bin is reaching full capacity. This enables timely human intervention only when required, thereby avoiding spillage and unnecessary checking.

Figure 3.2 illustrates this end-to-end process, beginning with initial trash detection, followed by navigation and unloading, and concluding with final classification and notification. This structured workflow illustrates how multiple subsystems come together in unison to achieve the overall goal of autonomous waste management in railway stations, while ensuring crowd-aware safety, blueprint-based navigation accuracy, and efficient bin monitoring.

3.3 Individual Component

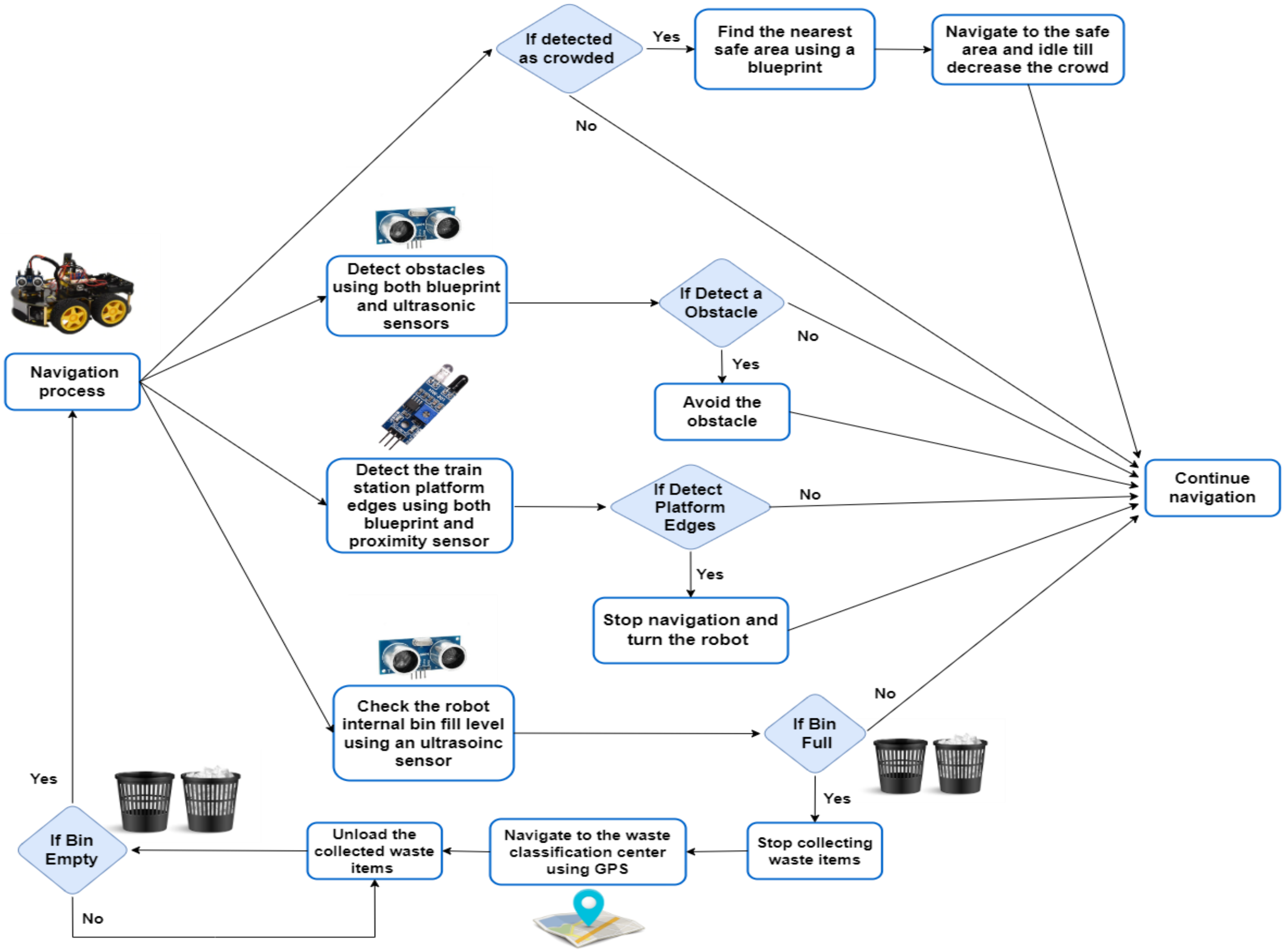

Figure 3.3: Individual component diagram

Figure 3.3 illustrates the architecture of the intelligent navigation and unloading control system, showing how obstacle avoidance, platform-edge protection, bin-level monitoring, and unloading mechanisms are integrated to enable safe and autonomous operation within railway stations. The intelligent navigation and unloading are main parts of the autonomous waste management system that enables the mobile robot to travel safely, effectively, and autonomously in the dynamic railway station environment. This sub-component manages two primary competencies: (i) autonomous navigation with strong obstacle avoidance and platform-edge safety, and (ii) intelligent unloading of the internal garbage bin into the right disposal station when full. Together, all these functions allow the robot to automatically control its collection cycle with no human aid while maintaining operational safety and efficiency.

The operation of this unit begins with environmental sensing for navigation. The robot scans its environment periodically using short-range ultrasonic sensors at various heights. They allow it to differentiate between low-level waste objects, floor clutter, and large objects such as benches, walls, or human legs. Proximity sensors protects against detection of sudden drop-offs such as platform edges to prevent accidents such as falling on railway tracks. In addition to awareness based on sensors, the robot also has blueprint-based spatial knowledge of the station layout, which enables it to recognize pre-defined static obstacles, edges, and safe regions. This combination of real-time sensing and blueprint-awareness supports more robust navigation in semi-structured railway environments.

Upon sensing the environment, navigation logic interprets the information in real time to control the motion of the robot. Instead of employing high-level path-planning algorithms, the system employs rule-based decision logic optimized for very structured station environments. For example, whenever there is a sensed obstacle within a threshold from the robot, the robot halts, calculates a new short movement vector, and resumes scanning for waste. If a sudden increase in detected head count indicates crowd congestion, the robot safely moves to the nearest predefined safe zone from the blueprint and pauses its task until the density decreases, thereby avoiding collisions. This low-weight navigation approach is computationally light and feasible in the intended environment, where flexibility and avoidance of obstacles are prioritized over global path optimization.

In parallel to navigation, the internal bin monitoring subsystem avoids the robot operating beyond its storage capacity. An ultrasonic sensor that is mounted inside the onboard bin constantly monitors the level of waste by estimating the remaining empty distance. These readings are converted to a fill level, and once the trigger (e.g., 80% full) is reached, the robot switches from "collection mode" to "unloading mode." This is an intelligent decision that plays a crucial role in making the operation smooth.

Location-awareness initiates unloading. An odometry and inertial measurement supported GPS module navigates the robot to the programmed point of disposal. Since train stations have partially covered platforms on which GPS performance is uncertain, the navigation is validated by sensor fusion combining location data with proximity sensing to adjust positioning as one approaches the unloading station. Upon reaching the disposal platform, the unloading system takes over. A servo-controlled bin flap or lifting system discharges the content onto the disposal platform in a way that the transfer is smooth without spilling any of the content.

Following unloading, robot logic reinitializes its state and resumes collection mode to return to dirty spots in the station. This continuous loop generates ongoing cleaning without human intervention. Putting obstacle avoidance, platform protection, bin monitoring, blueprint-based navigation, crowd-aware safety, and unloading into a single control unit gives an entire autonomous waste management cycle.

The novelty of this module is that it combines navigation and unloading capabilities optimized for railway environments. Most robotic platforms focus on just cleaning or collection, but this system acutely combines real-time obstacle avoidance, edge platform safety, internal bin monitoring, GPS-based localization-intelligent unloading, blueprint-driven safe zone handling, and crowd-adaptive behavior. Its focus on light-weight navigation without relying on heavy computational algorithms or cloud-based control also makes it realistic and scalable in actual applications.

By combining optimal unloading and safety-conscious navigation into one process, this module enables the robot to operate independently in crowded, complex railway stations. Not only does the resulting system save human labor and operational inefficiencies, but it also demonstrates how intelligence-enhanced IoT-powered robotics can address an immediate public sanitation problem on Sri Lanka's railway station infrastructure.

3.4 Project Management and Development Approach

3.4.1 Requirements gathering and analysis

This phase is aimed at establishing specific requirements of the system via analysis of the waste problem at railway stations. It involves communication with stakeholders (e.g., railway staff, cleaners, and supervisors), literature analysis, and mapping system requirements to functional and non-functional specifications. The outcome is an evident and verified set of requirements that guide design and implementation.

3.4.2 Technology selection

Here, the best hardware (e.g., microcontrollers, sensors, robot parts) and software (machine learning systems, IoT platforms, mobile application technologies) are chosen. They are chosen on the basis of scalability, affordability, reliability under railway conditions, and simplicity of integration. This makes the project include optimized, pragmatic, and green technologies.

3.4.3 Agile sprint planning

The project is then divided into smaller, bite-sized chunks referred to as sprints. Every sprint is of a defined duration (e.g., 2–3 weeks) and is designed to release a particular set of features, such as garbage detection, navigation, or unloading. Tasks are delegated, prioritized, and scheduled within sprint planning to ensure there is an even workload distribution for the team.

3.4.4 Agile sprint execution

This stage is the coding and rollout of planned features. Members of the team write code, test, and integrate their pieces continuously collaborating. Daily or weekly stand-ups ensure early detection of blockers, and tracking is done closely. The rollout is incremental and hence changes and feedback get embedded within the sprint cycle.

3.4.5 Agile sprint review

At the end of each sprint, review meetings are organized in which delivered functionality is presented to supervisors and colleagues. Feedback is obtained in order to have the deliverables aligned with expectations and tasks designed for subsequent sprints. This keeps stakeholders involved and quality assured throughout the project.

3.4.6 Deployment and maintenance

Once core development is complete, the system is installed in a test environment that simulates railway stations. Maintenance processes are implemented to handle updates, bug fixing, and real-world performance improvement. This stage ensures system stability and flexibility over first-time deployment.

3.4.7 Quality assurance

Quality assurance (QA) runs throughout all phases, not just at the end. Rigorous testing ensures reliability in navigation, waste detection, classification, and communication with the cloud. Performance tests, safety validation (e.g., fall detection at platforms), and accuracy of machine learning models are included to ensure the system achieves its objectives.

3.5 Hardware, Software, and Resource Methodology

3.5.1 Hardware (IoT and robotics hardware)

The hardware side enables the robot to sense, interact, and act upon the railway station surroundings. Each device is selected based on cost-effectiveness, ruggedness, robustness, and suitability for practical implementation.

- **Sensors**
    - **Ultrasonic Sensors:** Use to detect obstacles, waste/object differentiation, and prevent falling from high surfaces. Chosen because they are inexpensive, consistent in widely different lighting conditions, and easy to interface with ESP32.
    - **Internal Bin Ultrasonic Sensor:** Place inside the internal bin to measure fill levels in real time. Selected for simplicity in calculating distance measurement to percentage fill.
    - **GPS Module:** Selected for calculating disposal station position. Although GPS location accuracy is not always reliable, it is reliable enough for pre-specified disposal positions at railway stations.
    - **IMU + Odometry Encoders:** Use to sense robot orientation and movement. Chosen to complement GPS data for finer control of movement in partially covered areas.

- **Actuators**
  - **Servo Motors / Linear Actuators:** Use to drive unloading devices to tilt or open the internal bin. Chosen for precision and miniaturization in compact robotic units.
  - **DC Motors:** Drive the robot's wheels. These are chosen for their ruggedness and common availability in low-cost robotic platforms.
- **Chassis and structure**
  - A minimalist robot chassis for a mobile robot is selected for optimal weight and stability with bearing onboard sensors, motors, and internal bin. Facile modification is allowed by aluminum frame.

3.5.2 Computer support

Processing and decision-making rely on efficient embedded systems that avoid costly hardware but achieve enough computational power.

- **Microcontroller and onboard processor**
  - **ESP32:** Selected as the central processing unit since it is dual-core, contains Wi-Fi/Bluetooth onboard, and is capable of processing low-level sensor inputs as well as logic-control at high levels. ESP32 is considerably cheaper and power-hungry than Raspberry Pi and can be used in battery-powered mobile robots.
- **Cloud support**
  - **Backend: Express js + Node js**
    Selected due to ease of integration with IoT devices and rapid deployment of APIs.
  - **Database: MongoDB**
    Used as a NoSQL database to hold bin level logs and GPS navigation information. Its flexibility enables real-time updating with effectiveness.
  - **Hosting: AWS**
    Enables remote monitoring, mobile app integration, and push notifications for bin status.

- **Development machines**
  - Generic laptops/PCs will be used during development for programming, debugging, and model training. Picked due to the availability of essential computational resources for development without costing too much for the project.

3.5.3 Supporting resources

Supporting resources offers connectivity, stability, and collaborative development.

- **Communication modules**
  - **Wi-Fi (ESP32 onboard) and Bluetooth:** Chosen for low-power, secure data transfer to the cloud and mobile applications.
- **Power management**
  - **Rechargeable Li-ion Battery Packs with Controllers:** Chosen to offer high energy density and reusability so the robot operates for extended amounts of time without the need for continuous recharging.
- **Software tools**
  - **Arduino IDE:** Chosen for microcontroller programming since it is user-friendly and offers comprehensive library support.
- **Collaboration tools**
  - **GitHub:** Chosen for team collaboration and version control.
  - **MS Planner:** Used for agile project management so that tasks are tracked and deadlines met.

3.6 Data Requirements and Collection Plan

Data requirement for this module is required in order to enable the mobile robot to safely move, navigate through obstacles, monitor its internal garbage bin, and unload waste properly at the disposal site. The data requirement comes mainly in two types: **navigation and obstacle avoidance data**, and **bin monitoring and unloading data.**

3.6.1 Navigation and obstacle avoidance data

In order for safe navigation to be provided within the railway stations, the robot relies on live data from a variety of sensors.

- Front and side-mounted ultrasonic sensors provide distance measurement for detecting and avoiding stationary or moving objects.
- Proximity sensors aid in the detection of sudden platform edges to avoid the robot falling off.
- GPS location data is utilized for identification and designation of disposal location so that the robot can be instructed on how to return to the correct unloading point after filling the internal bin.

Navigation data will be collected during controlled testing, such as running the robot through high-obstacle terrain and testing near platform-like edge areas. These tests will validate detection precision, response times, and safe stopping behavior.

3.6.2 Bin fill level and unloading data

Unloading must be overdone with a check on the internal garbage bin capacity.

- An ultrasonic sensor inside the internal bin that is constantly checking the level of waste.
- Test data points such as "empty," "partially full," and "full" will be used for testing to ensure proper detection of thresholds.
- The sequence of unloading will be done and tested after the bin has reached across the threshold (e.g., 80%).

Data regarding unloading will be gathered utilizing repeated load–unload simulation to gather data on repeated loading and unloading, bin opening mechanism reliability testing, and waste proper deposition in the dumping area.

3.6.3 Data collection method

The data will be acquired primarily through repeated physical testing in test environment hardware and station-like settings. Each test trial will produce quantifiable data such as sensor readings, navigation logs, and unloading failure/success notifications. The data will be recorded digitally and analyzed to enhance the obstacle avoidance reasoning, increase the accuracy of bin status monitoring, and optimize unloading processes.

3.7 Gantt Chart

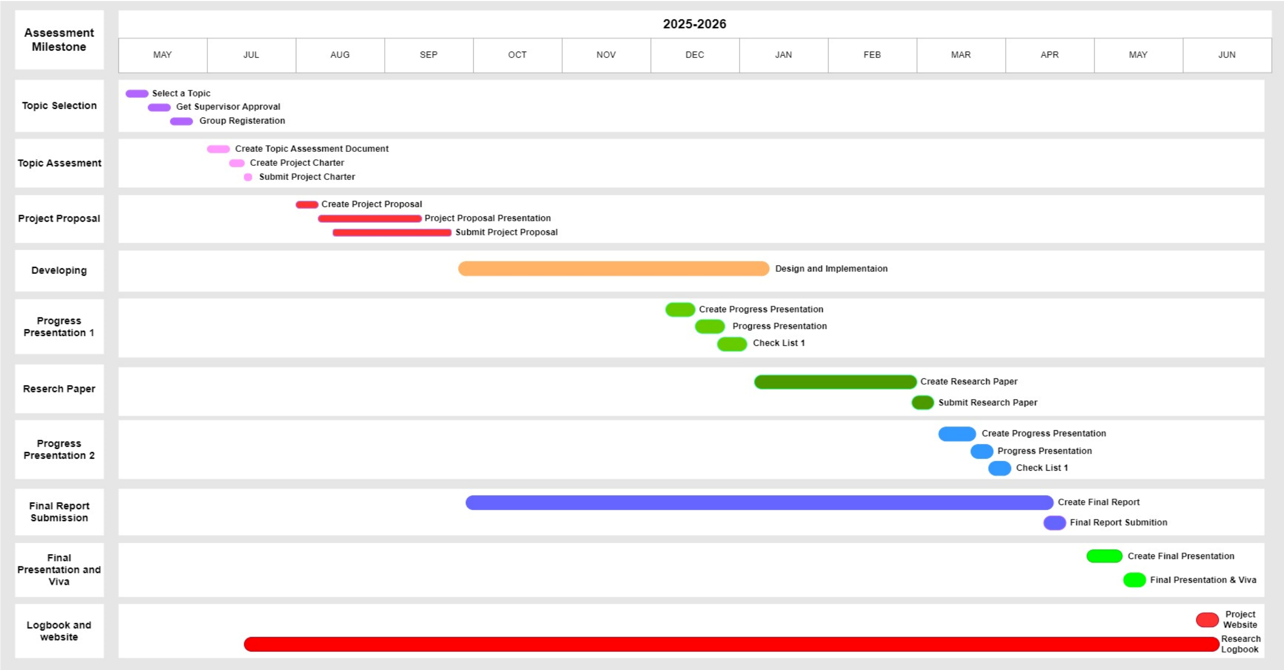

Figure 3.4: Gantt chart

3.8 Work Breakdown Chart

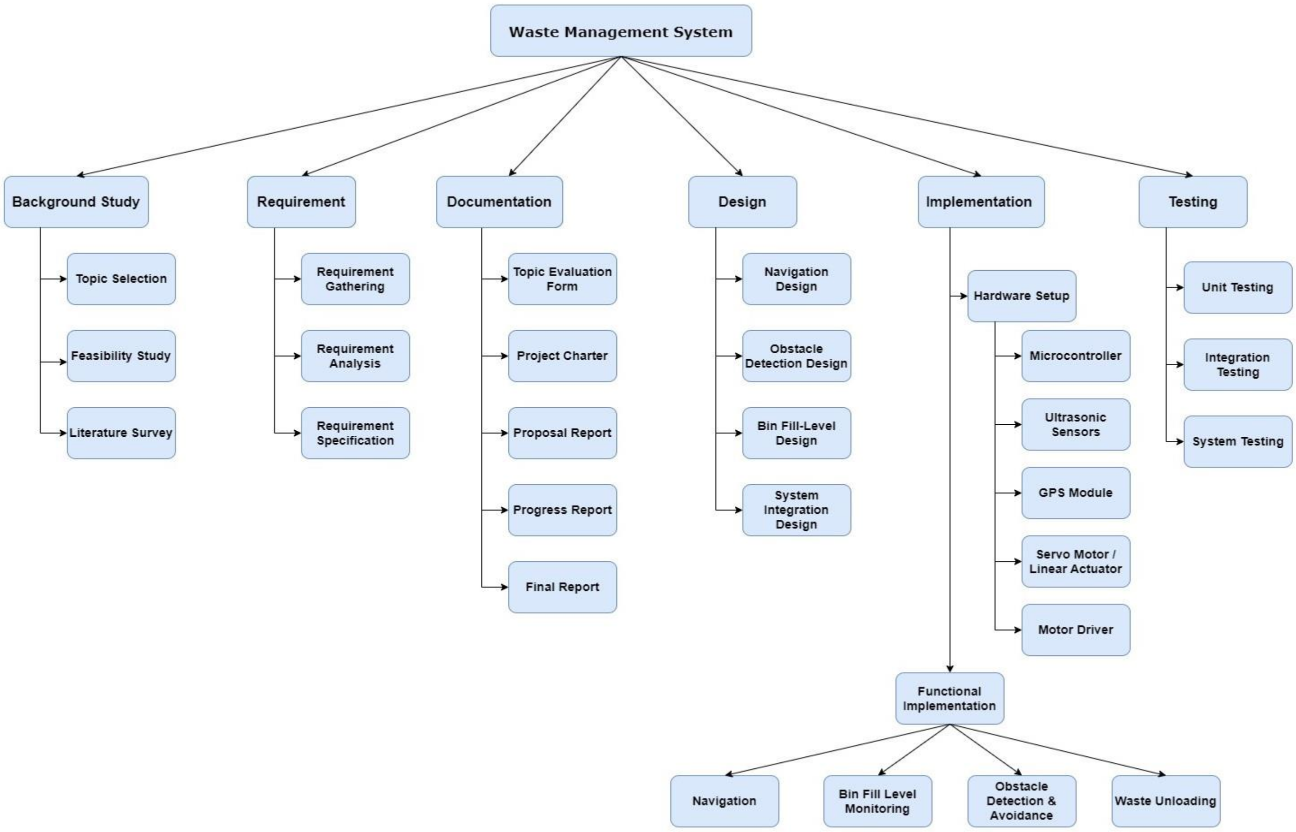

Figure 3.5: Work breakdown chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The robot should autonomously navigate railway station environments while avoiding static and dynamic obstacles using onboard sensors (ultrasonic).
- The robot should detect platform edges and prevent itself from falling using downward-facing sensors.
- The robot should monitor the internal trash can fill level with sensors.
- The robot should detect dumping stations with GPS/location awareness and stop automatically for unloading.
- The robot should only initiate unloading when the bin is full or at a staged disposal point.

4.2 Non-functional Requirements

- **Performance:** The system must process sensor data and execute navigation or unloading decisions in real time with minimal latency.
- **Scalability:** The design should support integration with additional sensors, extended routes, or larger waste bins without major modifications.
- **Reliability:** The navigation and unloading functions must operate consistently under varying conditions to minimize failures or downtime.
- **Security:** All data transmitted between the robot and monitoring systems must be protected from unauthorized access or tampering.
- **Usability:** The system should be intuitive for operators, requiring minimal training while providing clear alerts and status updates.

4.3 User Requirements

- The robot should unload collected garbage into the correct waste disposal location without any human intervention.
- The system should unload only when the internal bin is full in an effort to reduce unnecessary unloading trips.

- Unloading and navigation should be safe for passengers and crew, avoiding collision and ensuring the robot does not tip over from the railway platform.
- The system will minimize disruption to passenger flow by operating efficiently and unobtrusively in congested environments.

4.4 System Requirements

- **Hardware:**
  - Ultrasonic sensors for obstacle avoidance and edge detection.
  - Proximity sensors for platform drop-off detection.
  - GPS/RTK-GPS module for disposal station localization.
  - Internal bin-level sensor (ultrasonic).
  - Actuator system for autonomous unloading.

- **Software:**
  - Embedded obstacle avoidance navigation logic.
  - Unloading decision algorithm from bin-fill data.
  - Real-time monitoring via cloud integration is provided.

5. DESCRIPTION OF PERSONAL AND FACILITIES

This component is carried out by Sendanayaka H.K.(IT22582324) under the guidance of supervisors. The responsibilities are designing and implementing the autonomous navigation process with obstacle avoidance, monitoring the internal bin's fill level, and controlling the unloading operation at the disposal area. The work is supported by facilities including access to IoT labs with mobile robot platforms, navigation sensors (ultrasonic/infrared), test areas replicating railway station environments, and software tools for integration and validation.

6. BUDGET AND BUDGET JUSTIFICATION

Table 6.1 Budget plan

Purpose	Estimated Cost (LKR)
Ultrasonic sensors (for obstacle avoidance & bin level detection - 4 to 8 units	960
Proximity sensors (for platform edge / fall detection) - 4 units	600
GPS module for disposal area identification	1100
Motor driver & actuators for unloading mechanism	1800
Microcontrollers (ESP32 and Arduino UNO) for navigation & control	4000
Power supply (Li-ion batteries, charging circuit)	4000
Wheel, tyre cuppling and bracket set - 4 unit	5600
Robot frame integration materials (mounts, wiring, casing)	5000
Total Estimated Cost	**23,060**

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